

Ex.no:7b

DATA VISUALIZATION USING TABLEAU - HOTEL BOOKING ANALYSIS

Aim:

To connect, analyze, visualize, and create interactive dashboards using Tableau for effective business intelligence and data analytics on hotel booking performance.

Algorithm:

Algorithm: Hotel Booking Analysis Visualization using Tableau

Step 1: Data Loading

Connect the synthetic Hotel Booking dataset to Tableau.

Verify the imported fields: Booking_ID, Room_Type, Booking_Channel, and Revenue.

The dataset contains room types (Deluxe, Standard, Suite) and booking channels (Direct, OTA, Corporate, Travel Agent).

Step 2: Room Type Revenue Bar Chart

Create a worksheet for room-wise revenue.

Place Room_Type on Rows and Revenue on Columns.

Select the Horizontal Bar Chart visualization.

Enable data labels to display the total revenue generated by each room type.

Step 3: Room Type Revenue Contribution

Create a worksheet using Room_Type and Revenue.

Aggregate Revenue using SUM.

Visualize the revenue contribution of each room type using a Pie Chart.

Display room type names and corresponding revenue as labels.

Step 4: Booking Channel Performance Trend

Create a Line Chart or Horizontal Bar Chart using Booking_Channel and Revenue.

Place Booking_Channel on Rows and Revenue on Columns.

Enable labels to display individual revenue figures.

Use the chart to compare performance across channels.

Step 5: Revenue Percentage Calculation

Create a calculated field named Percentage.

Use the formula:

$\text{SUM}([\text{Revenue}]) / \text{TOTAL}(\text{SUM}([\text{Revenue}])) * 100$

Display Booking_Channel, Revenue, and Percentage in a worksheet.

Step 6: Booking Channel Filter

Add Booking_Channel to the Filters shelf.

Select specific channels such as Direct or OTA.

Observe updated room revenues dynamically based on the selected channel.

Step 7: Minimum Revenue Parameter

Create a parameter named Minimum Revenue.

Set the required minimum revenue threshold.

Apply the parameter to display only room types or channels that meet or exceed the selected value (e.g., Minimum Revenue = 80K).

Step 8: Dashboard Creation

Create a new Tableau Dashboard.

Add the Room Type Revenue, Room Contribution, Revenue Percentage, Channel Performance Trend, and Minimum Revenue visualizations.

Arrange the worksheets clearly and format with suitable titles and labels.

Step 9: Story Creation

Create a Tableau Story using the completed dashboard.

Add story points for:

1. Overall Hotel Booking Performance
2. Channel-wise Breakdown
3. High-Performing Room Categories
4. Final Revenue Analysis

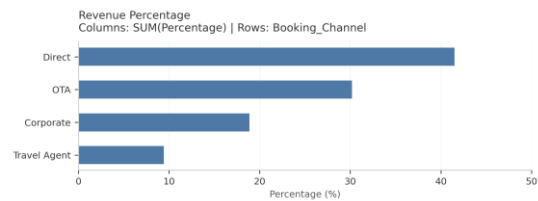
Step 10: Analysis

Analyze charts to identify top revenue-generating room types and booking channels.

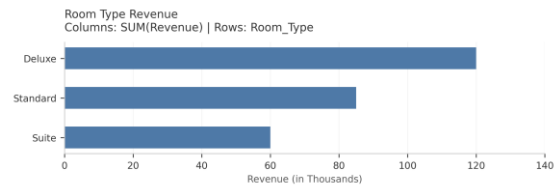
Observe how interactive filters and parameters update visualizations in real time.

Output:

Columns: SUM(Percentage) | Rows: Booking_Channel



Columns: SUM(Revenue) | Rows: Room_Type



Columns: SUM(Revenue) | Rows: Booking_Channel

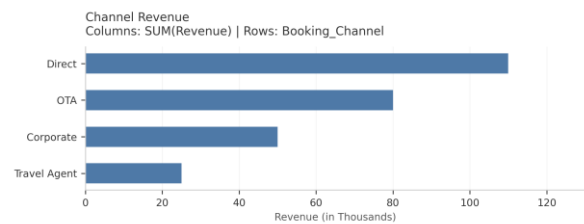


Tableau Dashboard & Interactive Visualizations



Result:

Data visualization using Tableau for the Hotel Booking dataset was successfully executed, creating interactive dashboards and story points for effective business insights.